

PANDAS

Cheat Sheet



1. Basic Commands

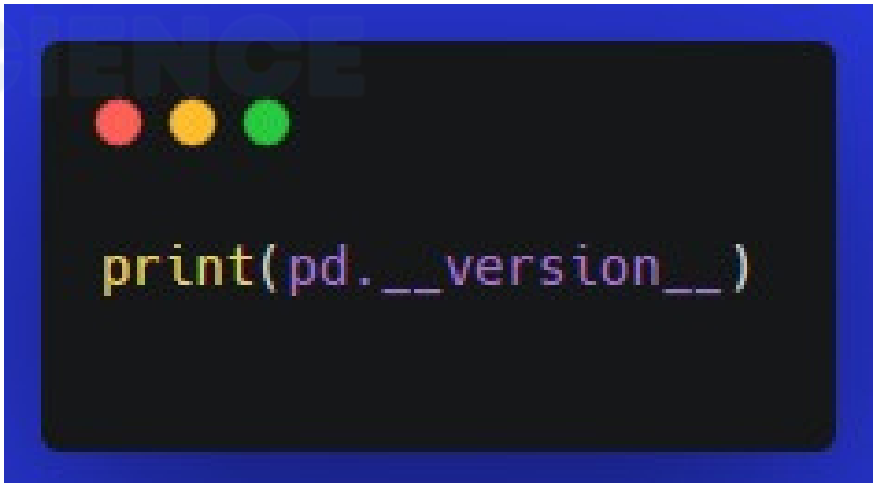
Pandas is a software library for Python that provides tools for data manipulation and analysis. It's important to ensure that the correct version of pandas is installed for compatibility with your code.

- Importing Pandas:



```
import pandas as pd
```

- Checking Pandas Version:



```
print(pd.__version__)
```



2. Dataframe Creation

Dataframes are two-dimensional labeled data structures with columns potentially of different types.

You can think of it like a spreadsheet or SQL table.

- From a list:

```
my_list = [1, 2, 3, 4, 5]
df = pd.DataFrame(my_list, columns=['column_name'])
```

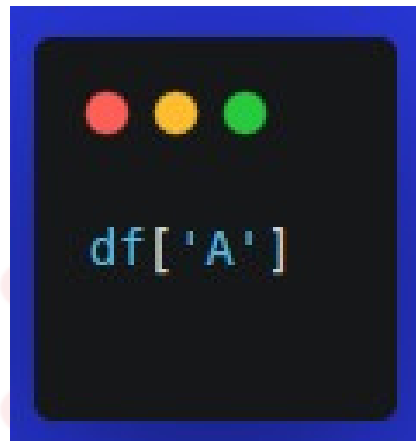
- From a Dictionary:

```
my_dict = {'A': [1, 2, 3], 'B': [4, 5, 6]}
df = pd.DataFrame(my_dict)
```



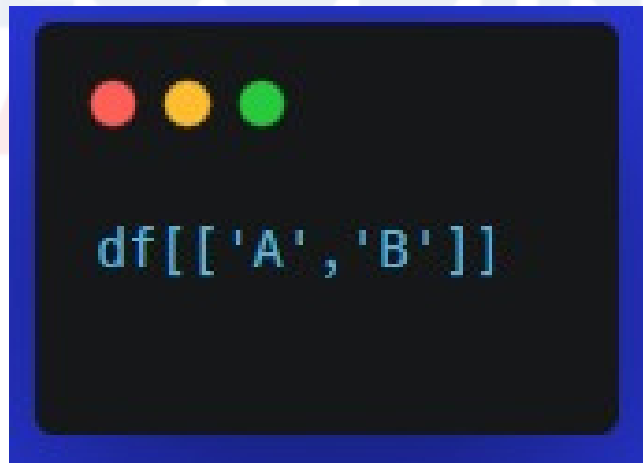
3. Data Selection

- Selecting a column:

A terminal window with a black background and a blue border. It features three colored circles (red, yellow, green) at the top. The code `df['A']` is displayed in a light blue font.

```
df['A']
```

- Selecting multiple columns:

A terminal window with a black background and a blue border. It features three colored circles (red, yellow, green) at the top. The code `df[['A', 'B']]` is displayed in a light blue font.

```
df[['A', 'B']]
```

Pandas provides different methods for data selection.



- Selecting rows:

```
df.loc[0] # row label  
df.iloc[0] # row index
```

- Selecting specific value:

```
df.at[0, 'A'] # row label and column name df.  
iat[0, 0] # row index and column index
```



4. Data Manipulation

Pandas provide various ways to manipulate a dataset.

- Adding a column:

```
df['C'] = pd.Series([7, 8, 9])
```

- Deleting a column:

```
df.drop('C', axis=1, inplace=True)
```



- Renaming columns:



```
df.rename(columns={'A': 'new_A'}, inplace=True)
```

- Applying a function to a column:



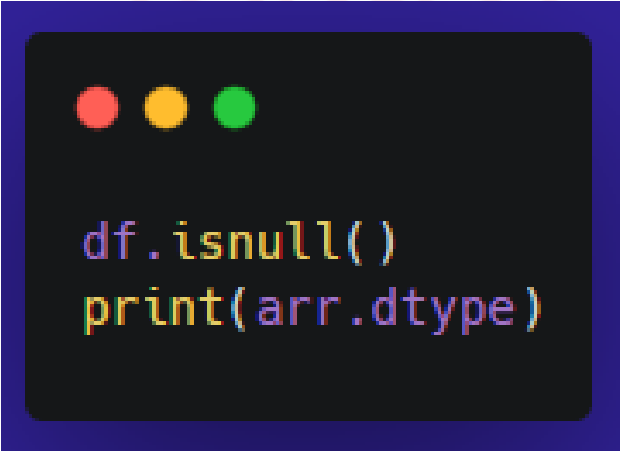
```
df['A'].apply(lambda x: x*2)
```



5. Data Cleaning

Data cleaning is detecting and correcting (or removing) corrupt or inaccurate records from a dataset.

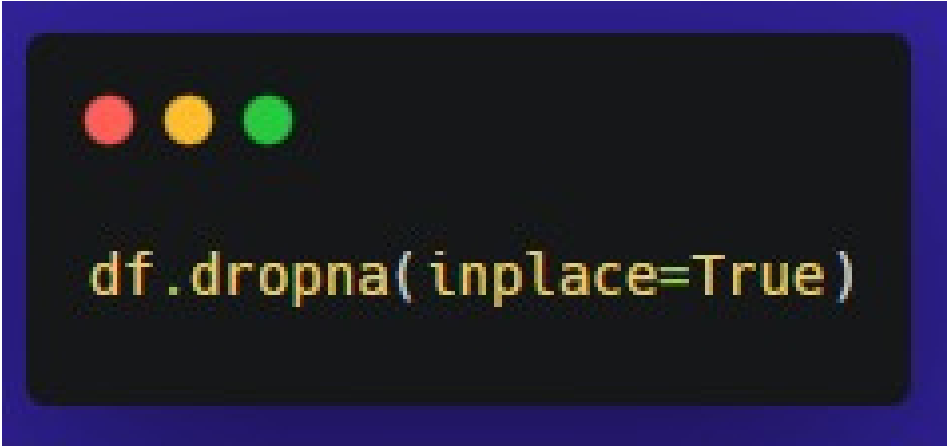
- Checking for null values:

A terminal window with a dark background and a blue border. It has three colored circles (red, yellow, green) in the top left corner. The code inside is:

```
df.isnull()  
print(arr.dtype)
```

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df.isnull()  
print(arr.dtype)
```

- Dropping null values:

A terminal window with a dark background and a blue border. It has three colored circles (red, yellow, green) in the top left corner. The code inside is:

```
df.dropna(inplace=True)
```

```
df.dropna(inplace=True)
```



Filling null values:



```
df.fillna(value=0, inplace=True)
```

- Replacing values:



```
df.replace(1, 10, inplace=True)
```



6. Grouping & Aggregation

Grouping involves combining data based on some criteria, while aggregation is the process of turning the results of a query into a single row.

- Group by:

```
df.groupby('A')
```

- Aggregation:

```
df.agg({'A': ['min', 'max', 'mean', 'sum']})
```



7. Merging, Joining, and Concatenating

Pandas provides various ways to combine DataFrames including merge and join.

- Concatenating:



```
df1 = pd.DataFrame({'A': [1, 2, 3], 'B': [4, 5, 6]})
df2 = pd.DataFrame({'A': [7, 8, 9], 'B': [10, 11, 12]})
df = pd.concat([df1, df2])
```



- Merging:

```
df1 = pd.DataFrame({'A': [1, 2, 3], 'B': [4, 5, 6]})  
df2 = pd.DataFrame({'A': [1, 2, 3], 'C': [7, 8, 9]})  
df = pd.merge(df1, df2, on='A')
```

- Joining:

```
df1 = pd.DataFrame({'A': [1, 2, 3], 'B': [4, 5, 6]})  
df2 = pd.DataFrame({'C': [7, 8, 9]})  
df = df1.join(df2)
```



8. Working with Dates

Pandas provides powerful functionalities for working with dates.

- Convert to datetime:

```
df['date'] = pd.to_datetime(df['date'])
```

- Extracting date parts:

```
df['year'] = df['date'].dt.year  
df['month'] = df['date'].dt.month  
df['day'] = df['date'].dt.day
```



9. File I/O

Pandas can seamlessly read from and write to a variety of file formats.

- Reading a CSV file:

```
df = pd.read_csv('file.csv')
```

- Writing to a CSV file:

```
df.to_csv('file.csv', index=False)
```

- Similarly for other file formats like

```
Excel (read_excel, to_excel), JSON (read_json, to_json), SQL (read_sql, to_sql), etc.
```



Next Steps

1. **Advanced Visualization Workshops:** Dive deeper into the intricacies of data visualization with hands-on workshops.
2. **Data Science Bootcamps:** A comprehensive, project-based learning experience to enhance your data science skills.
3. **AI Innovation Hub:** Collaborate with fellow learners and industry experts on cutting-edge AI projects.



Next Steps

4. Community Webinars: Regular webinars on the latest trends, tools, and best practices in AI and Data Science.

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Community Profile



What Does The Community Provide?

Gen AI Courses

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- ✓ **Prompt Engineering for Developers**
- ✓ **Langchain for AI App Development**

Recordings

- ✓ **Outcome-based Workshops**
- ✓ **AI Community Meetup Recordings**
- ✓ **Python Projects Videos**
- ✓ **AI & DS Career & Learning Webinar Series**

Data Science Courses

- ✓ **Basic Excel For Data Science**
- ✓ **Basic SQL For AI/Data Science**
- ✓ **Basic Python for AI/Data Jobs**
- ✓ **Advanced Python for AI/DS Jobs**
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- ✓ **Machine Learning**
- ✓ **Knowledge Shorts**

Resources

- ✓ **Generative AI Resources**
- ✓ **Sample Datasets & Projects**
- ✓ **Sample Reviewed Resume**
- ✓ **Ready to use Resume Template**
- ✓ **Linkedin Profile Optimization**
- ✓ **Essential SQL Documents**
- ✓ **Essential Python Documents**
- ✓ **Machine Learning Documents**

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